

Computer Networks

Lecture 25 Network Layer

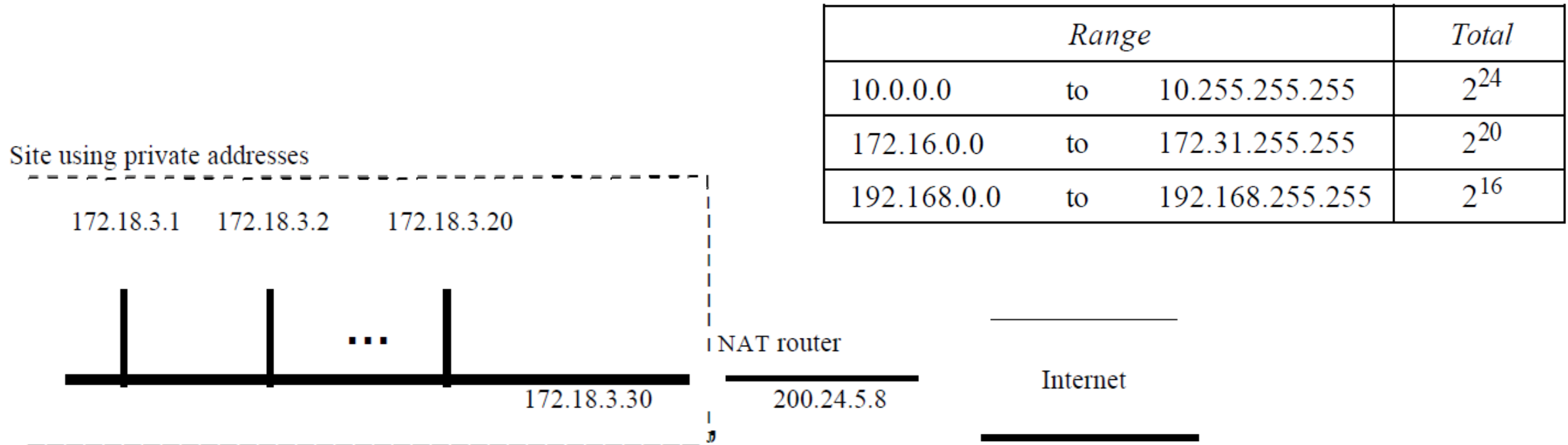
Content



- **NAT**
- **ARP**
- **RARP**
- **DHCP**
- **ICMP**
- **IGMP**
- **Routing Protocol**

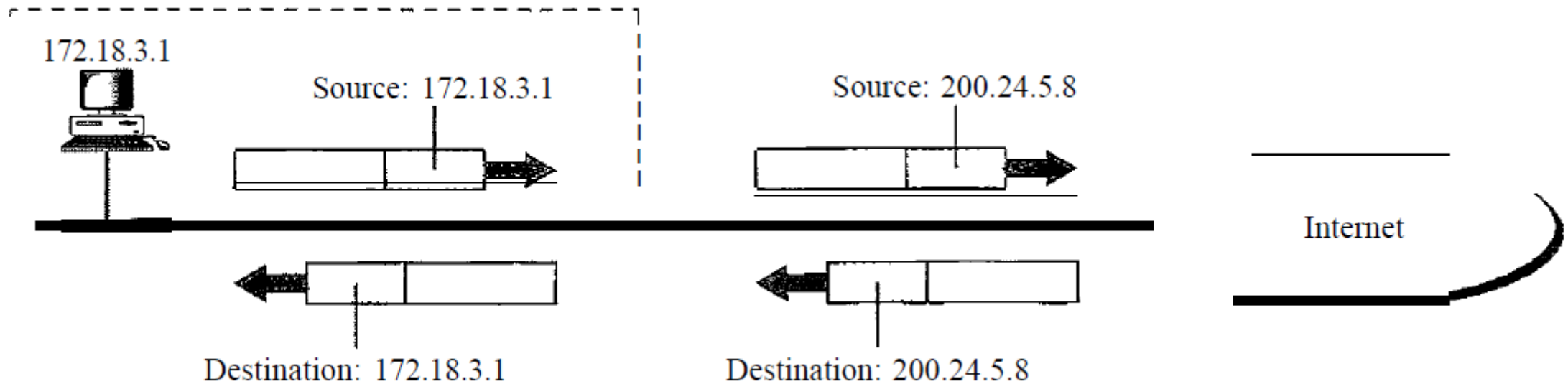
NAT

Network Address Translation (NAT) is a service that enables private IP networks to use the internet and cloud. NAT translates private IP addresses in an internal network to a public IP address before packets are sent to an external network.



NAT Address Translation

All the outgoing packets go through the NAT router, which replaces the *source address* in the packet with the global NAT address. All incoming packets also pass through the NAT router, which replaces the *destination address* in the packet with the appropriate private address.



NAT Translation Table

A NAT (Network Address Translation) translation table is a data structure maintained by a router or firewall that maps private (internal) IP addresses to public (external) IP addresses so that devices in a private network can communicate with external networks like the internet.

A typical NAT table entry includes:

- **Inside Local Address** → Private IP of the device inside the network
- **Inside Global Address** → Public IP assigned by NAT
- **Outside Local Address** → Destination IP as seen from inside
- **Outside Global Address** → Actual destination IP on the internet
- **Port Numbers** → Used in PAT (Port Address Translation) to distinguish sessions
- **Protocol** → TCP / UDP / ICMP

NAT Approach

1. A device inside the network (e.g., 192.168.1.10) sends a request to a server on the internet.
2. The NAT router:
 - Replaces the private IP with its public IP (e.g., 203.0.113.5)
 - Assigns a unique port (e.g., 3001)
3. It stores this mapping in the NAT table.
4. When the response comes back:
 - The router checks the table
 - Translates it back to the original private IP and port

Inside Local (Private)	Inside Global (Public)	Outside Local	Outside Global	Protocol
192.168.1.10:5000	203.0.113.5:3001	142.250.183.14:80	142.250.183.14:80	TCP
192.168.1.11:5001	203.0.113.5:3002	8.8.8.8:53	8.8.8.8:53	UDP

ARP Protocol

ARP (Address Resolution Protocol) is used in a local network to map an **IP address** → **MAC address**. In simple terms, it helps a device find the physical hardware address of another device when only the IP is known.

Devices communicate using:

- **IP address** → Logical (used for routing across networks)
- **MAC address** → Physical (used for communication within a LAN)

When a device wants to send data within the same network, it must know the destination MAC address, not just the IP. That's where ARP comes in.

ARP Protocol (Steps)

ARP Request (Broadcast)

- Sender asks:
"Who has IP 34.40.21.20? Tell me your MAC address."
- This message is broadcast to all devices in the LAN.

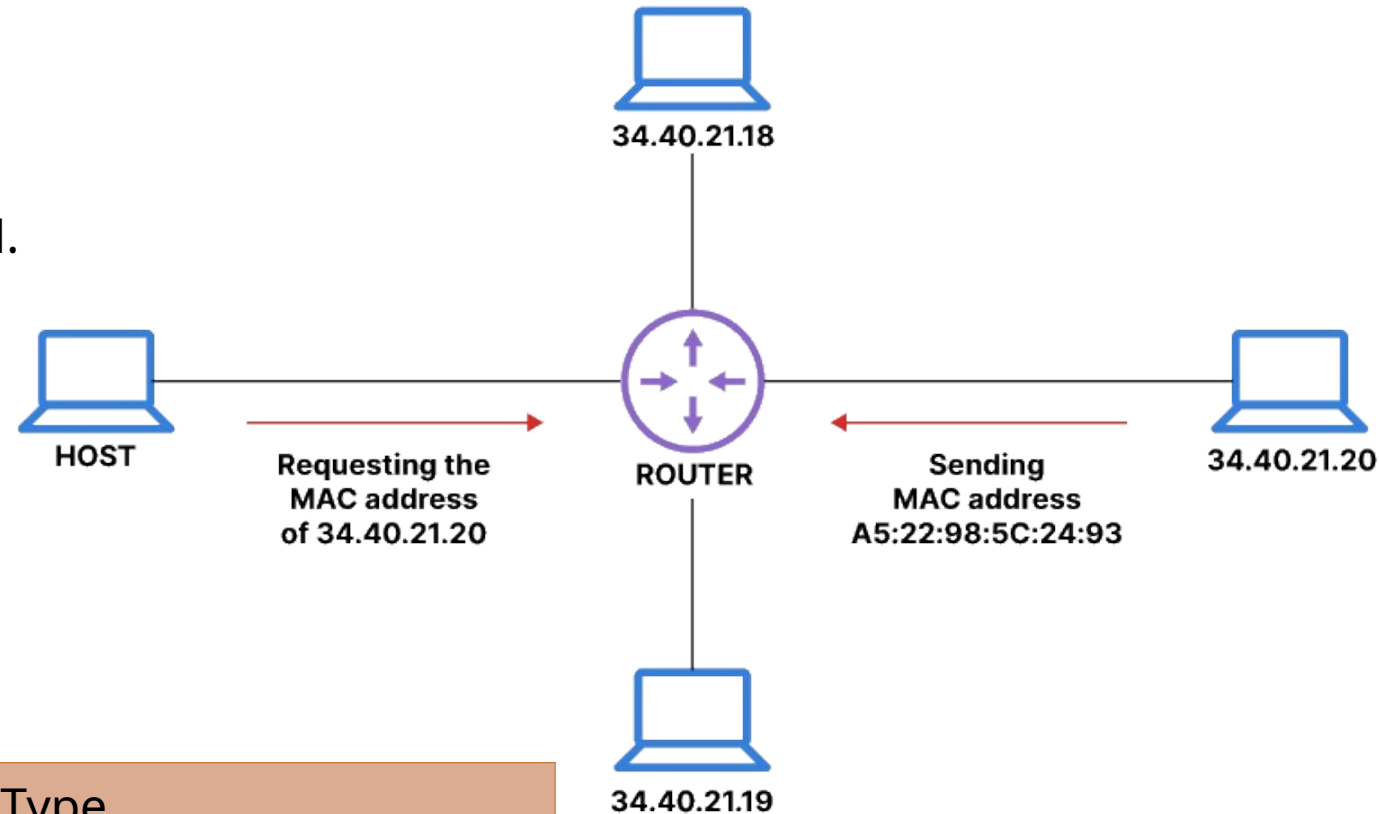
ARP Reply (Unicast)

- The device with that IP responds:
"I am 34.40.21.20, my MAC is A5:22:98:5C:24:93."

ARP Table Update

- The sender stores this mapping in its **ARP cache/table**.

IP Address	MAC Address	Type
192.168.1.1	AA:BB:CC:DD:EE:FF	Dynamic
192.168.1.10	11:22:33:44:55:66	Static



RARP Protocol

Some early devices (like diskless workstations) only knew their hardware (MAC) address but needed an IP address to communicate on the network.

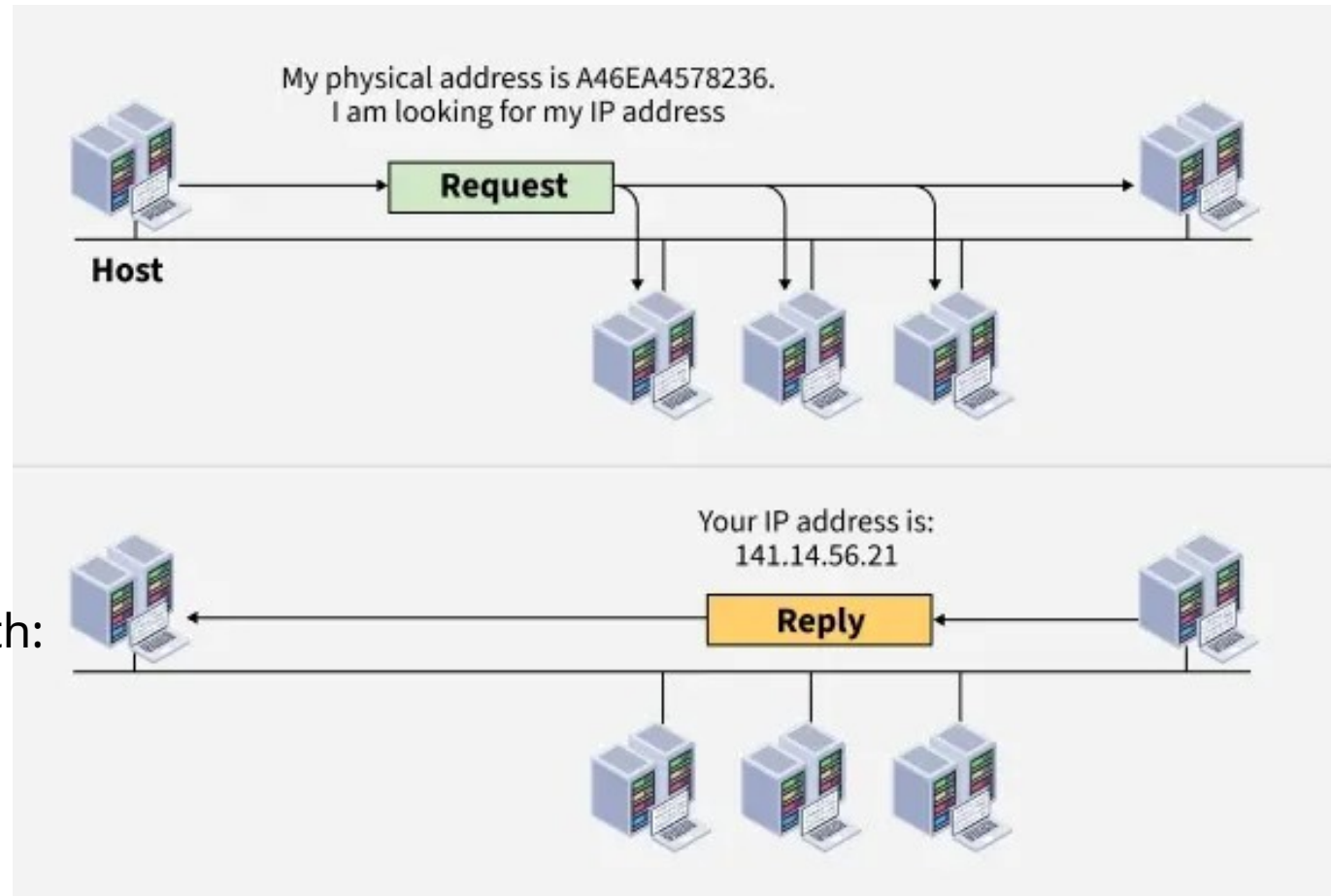
So they used RARP to ask:

“I know my MAC address — what is my IP address?”

RARP Protocol (Steps)

RARP Request (Broadcast)

- The device sends a broadcast:
"My MAC is A46EA4578236 — what is my IP?"



RARP Server Reply

- A RARP server looks up a table and responds with:
"Your IP address is 141.14.56.21."

RARP Protocol Limitations

- Required a dedicated RARP server
- Worked only within the same LAN (not routable)
- Provided only IP address, no other configuration (like gateway, DNS)

RARP is now obsolete and replaced by:

- BOOTP
- DHCP

These newer protocols provide:

- IP address
- Subnet mask
- Default gateway
- DNS servers

DHCP Protocol

DHCP is a network protocol that automatically assigns IP configuration to devices when they join a network.

When a device connects, DHCP assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server
- Lease time (how long the IP is valid)

DHCP Protocol (Steps)

Discover (Broadcast)

Client sends: *“Is there any DHCP server?”*

Offer

Server replies: *“I can give you this IP address.”*

Request

Client says: *“I accept this IP.”*

Acknowledge (ACK)

Server confirms and assigns configuration

DHCP Working Flow

- Initially, the client has **no IP**
- Uses broadcast to find DHCP server
- Gets full configuration in seconds
- Stores it temporarily (lease)

DHCP Protocol (Steps)

Case 1: DHCP Server in the Same LAN (Local)

- The DHCP server is inside your local network
- Client broadcasts (Discover) directly reach the server
- Simple setup, no extra configuration needed

Case 2: DHCP Server in a different network (Remote)

- The DHCP server is located in another subnet or data center
- Problem: DHCP Discover is a **broadcast**, and routers don't forward broadcasts

DHCP Protocol (Remote)

A router or Layer-3 switch acts as a relay:

- Client sends DHCP Discover (broadcast)
- Relay agent captures it
- Converts it to unicast and forwards to DHCP server
- Server replies → relay → client

DHCP Relay Agent

Enables communication between clients and remote DHCP servers

ICMP: Internet Control Message Protocol

ICMP is a network-layer protocol used for error reporting and diagnostics in IP networks. It helps devices understand what went wrong or whether a destination is reachable.

When packets travel across networks, things can go wrong:

- Destination unreachable
- Network congestion
- TTL expired

ICMP Steps

1. A device sends an IP packet
2. If an issue occurs:
 - Router/host generates an ICMP message
3. Message is sent back to the source

ICMP Messages

Echo Request / Echo Reply

- Used by the ping command
- Checks if a host is reachable

Destination Unreachable

- Sent when a packet cannot reach its destination
- Reasons:
 - Network unreachable
 - Host unreachable
 - Port unreachable

Time Exceeded

- Occurs when TTL (Time-To-Live) becomes zero
- Used in **traceroute** to track path

Redirect

- Router tells a host a better route exists

IGMP: Internet Group Message Protocol

IGMP is a network-layer protocol used to manage multicast group membership in IPv4 networks. It allows devices to join or leave multicast groups, so they receive only the data they are interested in.

- **Unicast** → one-to-one
- **Broadcast** → one-to-all
- **Multicast** → one-to-many (selected group)

IGMP helps manage this “selected group”.

IGMP: Working

1. Join Group

- A device sends an IGMP message:
“I want to join multicast group 224.0.0.1”

2. Router Tracks Membership

- Router keeps a record of which devices belong to which group

3. Multicast Data Delivery

- Router forwards multicast traffic only to group members

4. Leave Group

- Device sends a leave message when it no longer wants data

Routing

When a router receives a packet, its job is to forward it to the correct next network so it eventually reaches its destination. Since a router is connected to multiple networks, it must decide:

The “best” path is decided using a cost value stored in routing table.

A routing table can be static or dynamic.

- Static tables are manually configured,
- while dynamic tables are automatically updated when network conditions change.

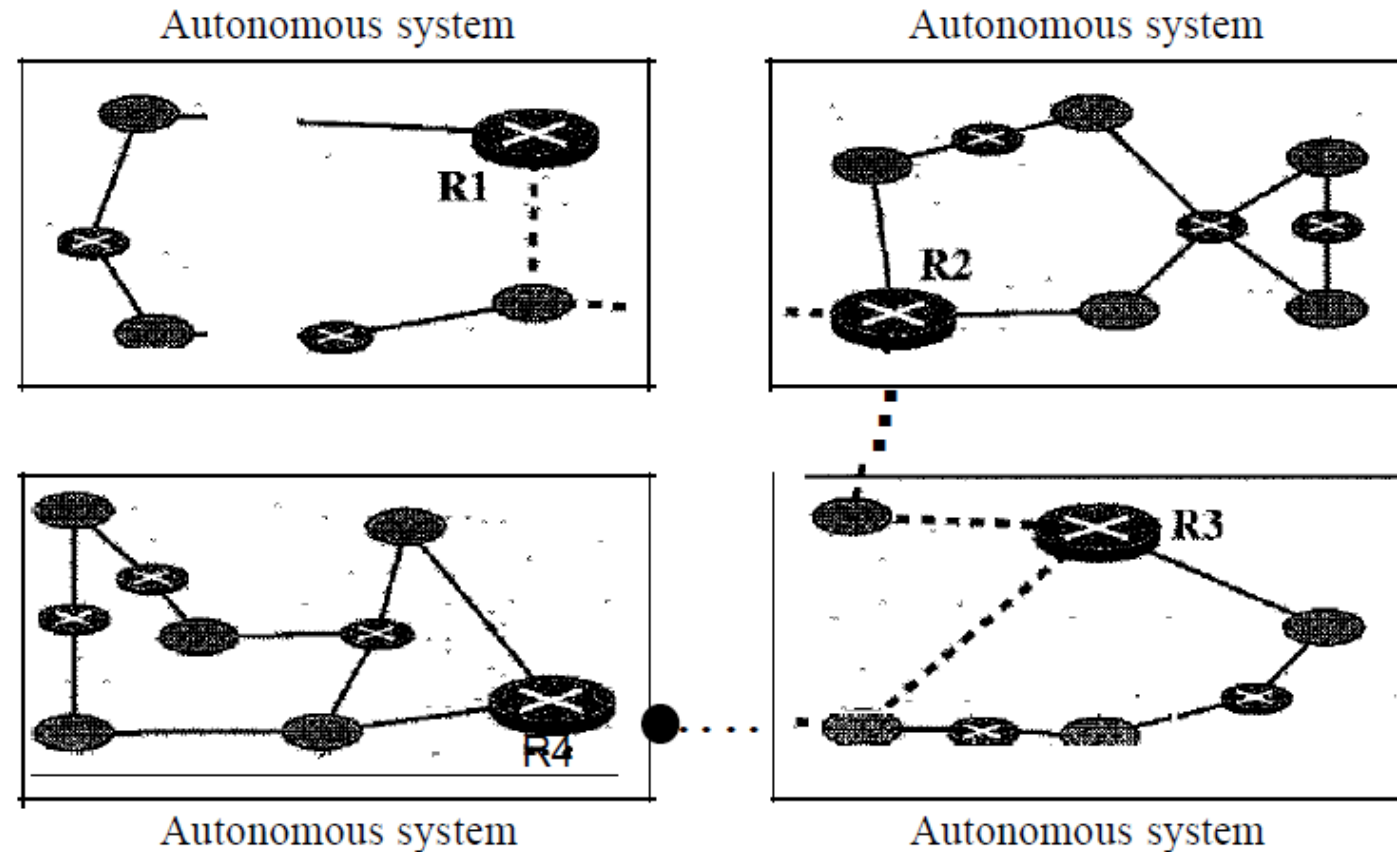
Modern networks rely on dynamic routing for accuracy and efficiency.

Routing Protocol enable routers to share information and update routes automatically, ensuring reliable communication.

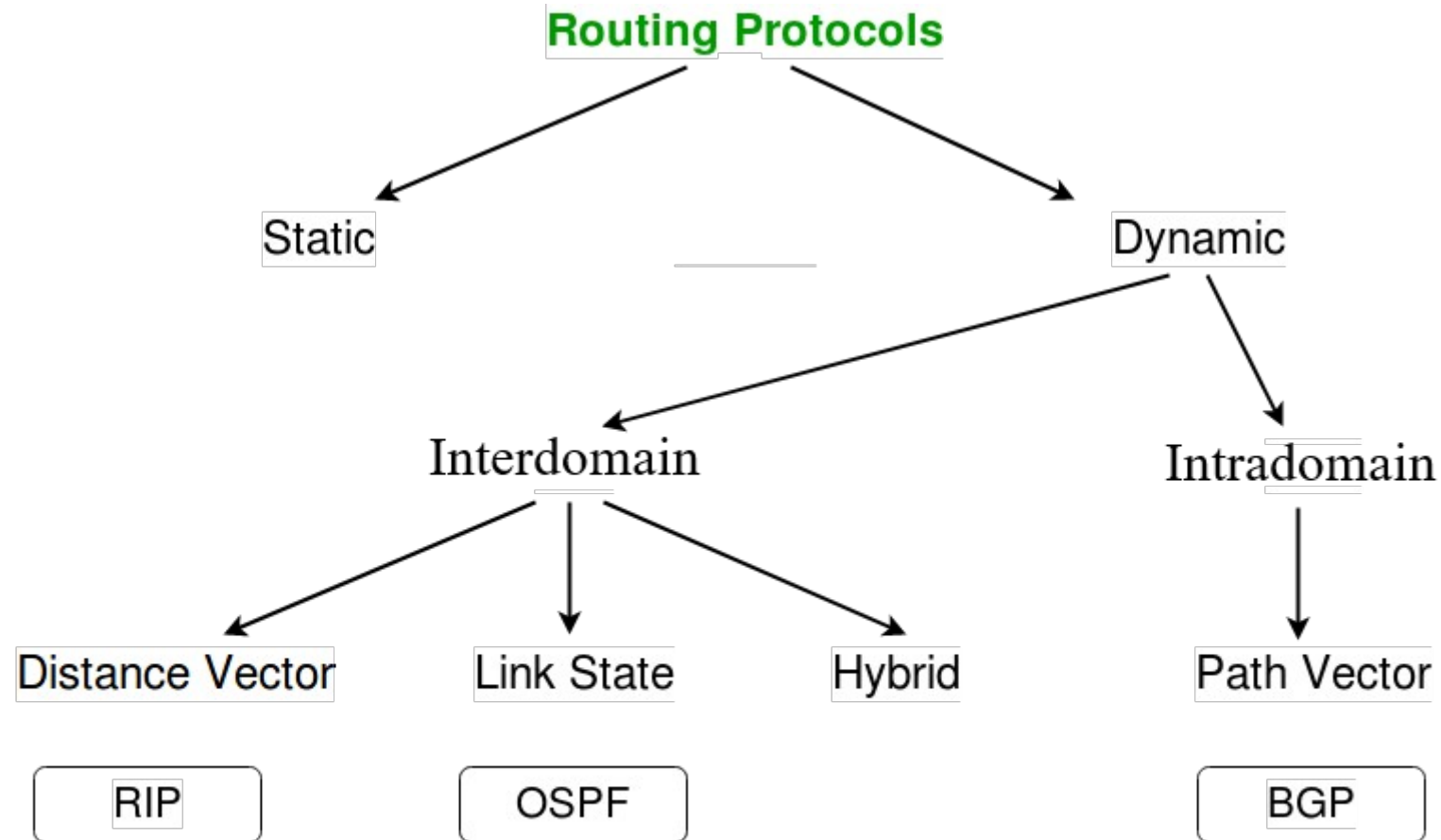
Routing :Intra and Inter Domain

In large networks, a single routing protocol cannot manage all routing updates efficiently. Therefore, the network is divided into Autonomous System (AS), which is a group of networks under one administration.

- Intradomain routing → Routing *within* an AS (uses protocols like OSPF, RIP)
- Interdomain routing → Routing *between* different ASes (handled by Border Gateway Protocol)



Routing :Intra and Inter Domain

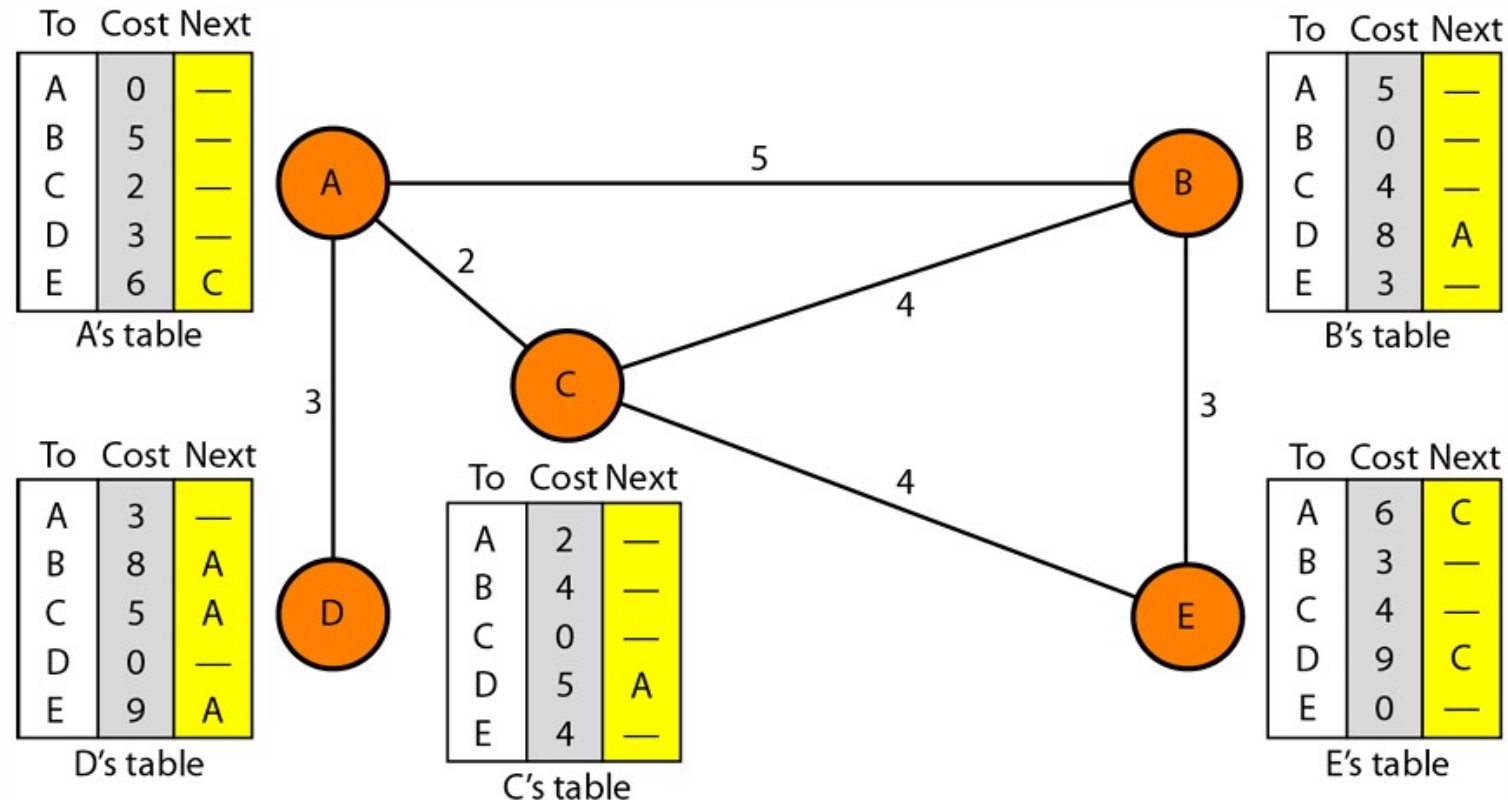


Distance Vector Routing

Distance Vector Routing is a routing method where each router determines the best path based on distance (cost)

In this protocol each node maintains a vector (table) of minimum distances to every node.

The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).

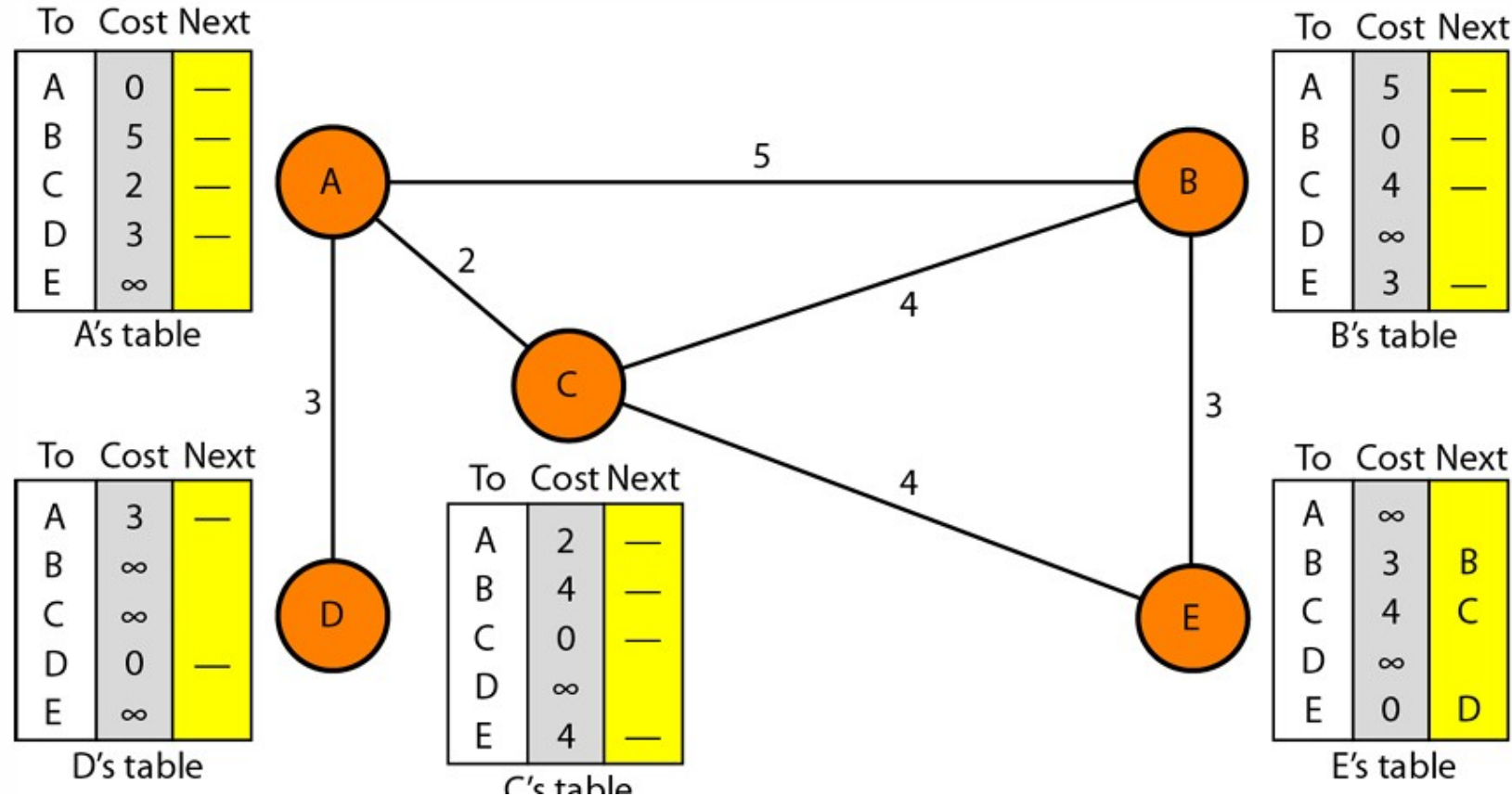


Distance Vector Routing : Initialization

The tables in last slide are stable, each node knows how to reach any other node and the cost.

At the beginning, however, this is not the case.

Each node can know only the distance between itself and its immediate neighbors.



Distance Vector Routing : Sharing

- Routers share routing information with their neighbors.
- A router can learn about unknown networks through neighbors
Example: A learns about E via C, and C learns about D via A
- Neighbors help each other improve routing tables

- Each router sends its routing table to neighbors
- Only first two columns are shared:
 - Destination
 - Distance (cost)

The next hop is not shared, because: Neighbor will replace it with the sender's address

Distance Vector Routing : Update

When a router receives a 2-column table (destination, cost) from a neighbor, it updates its own table.

- Step 1: Add the link cost (distance to neighbor) to each received entry
New cost = (neighbor's cost to destination) + (cost to reach neighbor)
- Step 2: For any usable entry,
set next hop = sender (neighbor) Because the route now goes via that neighbor
- Step 3: Compare with Existing Table For each destination:
 - Case 1: Different Next Hop Choose the route with lower cost
⇒ If costs are equal → keep the old route
 - Case 2: Same Next Hop Always update with new value Even if new cost is higher or infinity
Because it reflects the latest network condition
⇒ Old route may no longer exist

Distance Vector Routing : Update

